

15 Here are the first five terms of a quadratic sequence.

3 20 47 84 131

(a) Find an expression, in terms of n , for the n th term of this sequence.

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(3)

The terms of a different sequence are given by the rule $u_{n+1} = ku_n + k$ where k is a constant.

Given that $u_1 = 9$ and $u_2 = 4$

(b) find the value of u_4

$u_4 =$
(3)

(Total for Question 15 is 6 marks)